

LMS SaaS Platform

A full-stack multi-tenant Learning Management System with integrated CRM, Placement, Marketing Intelligence, and AI-powered features. Built for educational institutions, coding bootcamps, and professional training providers.

22

Modules

120+

Features

~65%

Overall Complete

8

User Roles



Completion Summary

5

Modules 85–95%

5

Modules 70–84%

7

Modules 50–69%

5

Modules <50%

#	Module	Completion	Progress	Status
1	Authentication & Onboarding	95%		✓ Production Ready
2	User Management	85%		✓ Stable
3	Roles & Permissions	85%		✓ Stable
4	Tenant Management	90%		✓ Production Ready
5	Course Management	80%		⚠ Minor gaps
6	Batch Management	75%		⚠ Reporting partial
7	Attendance System	60%		⚠ 15–20hrs work left
8	Quiz & Assessment	65%		⚠ 17–22hrs work left
9	Assignment Management	70%		⚠ Grading partial
10	Code Snippets	60%		⚠ No execution engine
11	Class Recordings	55%		⚠ ClassFlow incomplete
12	Mock Interviews	50%		⚠ AI eval missing
13	Leads & CRM	70%		⚠ Bulk import missing
14	Marketing Intelligence	40%		🚧 Early-stage
15	Placement & Alumni	45%		🚧 Core done
16	College Administration	50%		⚠ Hierarchy missing
17	Learning Hub	35%		🚧 Early-stage
18	Content Management	80%		✓ Stable
19	Notifications	50%		⚠ WS partial
20	Certificates	40%		🚧 Basic only
21	Dashboard & Analytics	65%		⚠ Partial analytics

#	Module	Completion	Progress	Status
22	System Administration	70%		⚠ Monitoring missing

Architecture & Dependencies

```

LMS SaaS Platform | |— Authentication (JWT + OAuth) | |— User Management
| | | |— Role & Permissions | | | |—
Tenant Management (Multi-tenant) | | |— Session / Refresh Token | | | |—
Learning Core | | |— Course Management |— All modules | | |— Batch
Management | scoped by | | | |— Attendance System | tenantId | | | |—
Class Recordings / ClassFlow | | | |— Content Management (WebSocket) | | |
| | | |— Enrollment | | | |— Assessments | | | |— Quiz & Assessment | | | |—
Question Bank | | | |— Assignment Management | | | |— Code Snippets | | | |—
Mock Interviews | | | |— Interview Question Bank | | | |— CRM & Sales | | |
| | | |— Leads & CRM | | | |— Lead Stages & Pipeline | | | |— Lead Scoring &
Priority | | | |— Telecaller Console | | | |— Follow-up Calendar | | |
| | | |— Lead AI Service (OpenAI GPT-4) | | | |— Google Sheet Integration | | |
| | | |— Marketing Intelligence | | | |— Ad Campaigns | | | |— Competitor
Tracking | | | |— Meta Webhook Integration | | | |— Sales Content Library
| | | |— Institution Management | | | |— College Administration | | | |—
Departments | | | |— Members | | | |— Bulk Upload | | | |— Placement &
Alumni | | | |— Placement Drives | | | |— Alumni Directory | | | |—
Student Applications | | | |— Learning Hub | | | |— Topic Mastery Heatmap |
| | | |— Learning Requests | | | |— Infrastructure |— Notifications
(WebSocket + Email) |— Certificates (Token-based) |— Dashboard &
Analytics |— System Administration | |— Feature Flags | |— Tenant
Settings |— File Upload Service (Multer)

```

1. Authentication & Onboarding



Authentication & Onboarding

Entry point for all users — login, registration, password management & OAuth

95% Complete

Public URL → `/login` or `/register` → JWT Token → `/dashboard`

Features

- ✓ Email/password login with JWT
- ✓ User registration (self-service)
- ✓ Organization creation at registration
- ✓ Forgot & Reset password flow
- ✓ Setup password (invite-based)
- ✓ Profile completion wizard
- ✓ OAuth (GitHub / LinkedIn) ready
- ✓ Refresh token mechanism
- ✓ bcrypt password hashing
- ⚠️ OAuth full flow (partial)

Routes

Path	Role
<code>/login</code>	Public
<code>/register</code>	Public
<code>/create-organization</code>	Public
<code>/forgot-password</code>	Public
<code>/reset-password</code>	Public
<code>/setup-password</code>	Invited
<code>/complete-profile</code>	Authenticated
<code>/profile/oauth-callback</code>	OAuth

Dependencies

None (Entry Point)

JWT Auth & Token Refresh		98%	✓
Password Management		95%	✓
OAuth Integration		60%	⚠
Profile Completion Wizard		92%	✓

2. User Management



User Management

Create, manage and monitor all users across the platform

85% Complete

Admin Sidebar → Users → /users → Create / Edit / Deactivate

- ✓ Full CRUD for users
- ✓ Role assignment per user
- ✓ User status (active / inactive)
- ✓ Bulk user operations
- ✓ Profile picture upload
- ✓ Search & filter
- ✓ Account deactivation
- ⚠ User activity tracking (partial)
- ✗ User analytics dashboard

Path	Role
/users	Admin
/profile	All
/admin/student-profiles	Admin
/admin/student-profiles/:id	Admin

Depends On

Authentication Roles Tenant Management

CRUD Operations		95%	✓
Invite & Onboarding		85%	✓
Activity Tracking		50%	⚠

3. Roles & Permissions



Roles & Permission Management

Fine-grained RBAC with 20+ permission types across all modules

85% Complete

Admin Sidebar → Roles → /roles → Create / Edit Role

Predefined Roles

- ✓ SUPER_ADMIN — full system access
- ✓ TENANT_ADMIN — tenant-level management
- ✓ INSTRUCTOR — course & content
- ✓ ATTENDANCE_ADMIN — attendance marking
- ✓ STAFF — sales, leads, enrollment
- ✓ STUDENT — learner access
- ✓ PLACEMENT_OFFICER — placement module
- ✓ GUEST — limited read access
- ✓ Custom roles
- ⚠ Permission inheritance (partial)

Key Permissions

manage_tenants · manage_all_users · manage_system_settings
manage_tenant_users · manage_roles · manage_tenant_settings
create_courses · manage_assignments · manage_interviews
mark_attendance · view_reports · export_leads
create_leads · manage_quizzes · view_analytics

Depends On

Authentication

User Management

4. Tenant Management



Tenant Management

Multi-tenant isolation, configuration, and subscription management

90% Complete

SUPER_ADMIN → Sidebar: Tenant Management → /super-admin/tenants

- ✓ Complete tenant isolation (middleware)
- ✓ Tenant CRUD by SUPER_ADMIN
- ✓ Multi-tenant API scoping
- ✓ Tenant-level configuration
- ✓ Subscription tier tracking
- ✓ Usage statistics per tenant
- ✓ Tenant-specific feature flags

Path	Role
/super-admin/tenants	SUPER_ADMIN

Architecture

Middleware-based tenant context injection. All DB queries filtered by `tenantId`. Cross-tenant access prevented at middleware layer.

5. Course Management



Course Management

Create, organize and manage courses with enrollments and instructor assignments

80% Complete

Admin Sidebar → Courses → /courses → Course Detail → Batches → Enrollment

- ✓ Create/edit/delete courses
- ✓ Course enrollment management
- ✓ Enrollment limits (capacity)
- ✓ Course status (active/draft/archived)
- ✓ Instructor assignment
- ✓ Course statistics
- ✓ Student course view (feature-gated)
- ⚠ Course content organization (via Content module)
- ✗ Course prerequisites
- ✗ Course templates

Path	Role
/courses	Admin/Instructor
/course-management	Admin/Instructor
/my-course	Student

Depends On

Authentication

User Management

Tenant Management

Required By

Batches

Attendance

Quizzes

Assignments

Content

Course CRUD		95%	✓
Enrollment System		85%	✓
Content Organization		60%	⚠
Prerequisites & Templates		0%	✗

6. Batch Management



Batch Management

Organize students into batches with schedules and capacity controls

75% Complete

Admin Sidebar → Batches → /batches

✓ Create/manage batches

✓ Student assignment to batches

✓ Batch capacity management

✓ Batch scheduling

✓ Batch statistics

⚠ Batch-level reporting (partial)

Path	Role
/batches	Admin/Instructor

Depends On

Courses

Users

7. Attendance System



Attendance System

Mark, track and report student attendance with time tracking

60% Complete

Admin Sidebar → Attendance → /attendance → Select Batch → Mark → Save

- ✓ Mark attendance (P/A/L status)
- ✓ In-time / out-time tracking
- ✓ Batch & student level views
- ✓ Attendance % calculation
- ✓ Date range filtering
- ✓ CSV export
- ⚠ Bulk marking (in progress)
- ⚠ Attendance notifications (TODO)
- ✗ QR code attendance
- ✗ Analytics dashboard

Path	Role
/attendance	Admin/Instructor
/my-attendance	Student
/attendance-reports	Admin

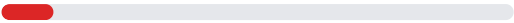
Depends On

Batches

Users

Courses

Basic Mark & Track		85%	✓
Reports & Export		70%	⚠
Bulk Operations		35%	⚠



8. Quiz & Assessment



Quiz & Assessment

Create question banks, configure quizzes with timers, and track student attempts

65% Complete

Admin → /quiz-management → Create Quiz → Link Questions → Publish →
 Student /quizzes → Take

- ✓ MCQ, Short Answer question types
- ✓ Question bank management
- ✓ Quiz creation wizard
- ✓ Question linking to quizzes
- ✓ Configurable timer
- ✓ Instant scoring
- ✓ Student attempt tracking
- ⚠ Auto-save answers
- ⚠ Question randomization (TODO 2hrs)
- ⚠ Answer review after submit (TODO)
- ✗ Quiz analytics dashboard
- ✗ Negative marking

Path	Role
/quizzes	Student
/quiz-management	Instructor
/question-bank	Instructor
/quiz/:id/take	Student
/quiz/:id/results/:attemptId	Student
/quiz-reports	Admin

Depends On

Courses

Users

Content Mgmt

Quiz Creation & Question Bank		85%	✓
Quiz Taking Interface		75%	⚠
Scoring & Results		70%	⚠
Analytics & Reports		35%	✗

9. Assignment Management



Assignment Management

Create assignments, manage submissions, and grade student work

70% Complete

Admin → /admin/assignments/create → Set Due Date → Student Submits →
Grade in /submissions

- ✓ Assignment CRUD
- ✓ Due date management
- ✓ File upload / download
- ✓ Submission tracking
- ✓ Grading interface
- ✓ Assignment reports
- ✗ Rubric-based grading
- ✗ Plagiarism detection

Path	Role
/assignments	Student
/admin/assignments	Admin
/admin/assignments/create	Admin
/admin/assignments/:id/submissions	Admin
/admin/assignments/reports	Admin
/assignments/:id/workspace	Student

Depends On

Courses Users

10. Code Snippets / Assessments



Code Snippets & Assessments

Coding assessments with submission tracking and grading

60% Complete

Admin → /admin/coding-snippets → Create Snippet → Student Submits → Grade

- ✓ Code snippet CRUD
- ✓ Submission grading
- ✓ Code syntax highlighting
- ⚠ Code execution engine (not implemented)
- ⚠ Test case validation (not implemented)

Path	Role
/admin/coding-snippets	Admin
/admin/coding-snippets/:id/submissions	Admin
/coding-snippets	Student

11. Class Recordings & Live Class



Class Recordings & Live Class

Record, store and serve class videos to students

55% Complete

Instructor → /admin/class-recordings/start → Upload Recording →

Student: /class-hub

- ✓ Recording upload & storage
- ✓ Recording playback
- ✓ Recording management
- ✓ Student recording access
- ⚠ ClassFlow unified interface (incomplete)
- ✗ Live video conferencing
- ✗ Recording transcription
- ✗ Chapter markers

Path	Role
/admin/class-recordings	Instructor
/admin/class-recordings/start	Instructor
/admin/class-flow	Instructor
/class-hub	Student
/class-recordings	Student

12. Mock Interviews



Mock Interviews

Create structured interview templates, assign to students, collect feedback

50% Complete

Admin → /admin/interviews/templates/create → Assign to Student → /student/interviews/take/:templateld → Analytics

- ✓ Interview template creation
- ✓ Question bank management
- ✓ Assignment to students
- ✓ Student interview taking interface
- ✓ Feedback report generation
- ✓ Analytics dashboard
- ⚠ AI-powered evaluation (not implemented)
- ✗ Video interview capability

Path	Role
/admin/interviews/templates	Admin
/admin/interviews/assignments	Admin
/admin/interviews/analytics	Admin
/student/interviews	Student
/student/interviews/take/:id	Student
/mock-interviews	Student

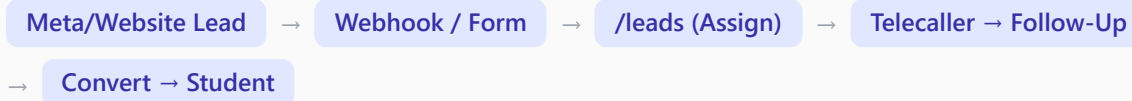
13. Leads & CRM System



Leads & CRM System

Full lead lifecycle management — capture, score, nurture and convert leads to students

70% Complete



✓ Lead creation (manual, webhook, form)

✓ Lead stages & Kanban pipeline

✓ Lead assignment to staff

✓ Follow-up tracking calendar

✓ Lead analytics & reports

✓ Manager performance board

✓ Telecaller console

✓ Lead priority (hot/warm/cold)

✓ Lead scoring rules engine

✓ Audit logs

✓ Form customization

✓ Google Sheets integration

✓ Meta webhook integration

✓ AI summary (OpenAI GPT-4)

⚠ Bulk lead import (missing)

⚠ Lead conversion tracking (partial)

Path	Role
<code>/leads</code>	Staff/Admin
<code>/leads/:leadId</code>	Staff
<code>/lead-stages</code>	Admin

/lead-form-settings	Admin
/lead-manager-board	Manager
/lead-my-performance	Staff
/lead-audit-logs	Admin
/telecaller-console	Staff
/lead-priority-settings	Admin
/qualification-settings	Admin
/sales-content	Staff
/leads/analytics	Admin
/follow-ups	Staff
/lead-scoring-settings	Admin
/google-sheet-integration	Admin

Depends On

- Users
- Roles
- Notifications

Integrations

- Meta/Facebook Webhook
- Google Sheets API
- OpenAI GPT-4

Lead Capture & Pipeline		85%	✓
Telecaller Console		80%	✓
Lead Scoring & Priority		75%	⚠
AI Summary (GPT-4)		60%	⚠
Bulk Import		0%	✗

14. Marketing Intelligence



Marketing Intelligence

Campaign tracking, competitor analysis, and AI-powered marketing content generation

40% Complete

Admin Sidebar → Marketing → /marketing

- ✓ Marketing dashboard
- ✓ Campaign management
- ✓ Competitor tracking
- ✓ Ad capture / monitoring
- ✓ Insights feed
- ⚠ AI content generation (partial)
- ✗ Social media integration
- ✗ Email campaign management

Path	Role
/marketing	Admin
/marketing/campaigns	Admin
/marketing/competitors	Admin
/marketing/ads	Admin
/marketing/insights	Admin

Depends On

- Leads CRM
- Meta Webhook

15. Placement & Alumni



Placement & Alumni

Manage placement drives, track student applications, maintain alumni directory

45% Complete

Placement Officer → /admin/college/placement → Create Drive →

Student: /student/my-applications

- ☒ Placement drive creation
- ☒ Placement analytics
- ☒ Alumni management
- ☒ Student application tracking
- ☒ Alumni directory
- ☐ ⚠ Job posting system (not documented)
- ☐ ✖ Offer management
- ☐ ✖ Salary tracking

Path	Role
/admin/college/placement	Placement Officer
/admin/college/placement-analytics	Admin
/admin/college/alumni	Admin
/student/my-applications	Student
/student/college	Student
/student/alumni-directory	Student

16. College Administration



College Administration

Manage departments, members, settings and bulk data operations

50% Complete

Admin Sidebar → College → /admin/college/departments

- ✓ Department CRUD
- ✓ Settings management
- ✓ Member management
- ✓ Bulk upload (CSV)
- ✓ College reports
- ⚠ Department hierarchy (not documented)
- ✗ Holiday calendar management

Path	Role
/admin/college/departments	Admin
/admin/college/settings	Admin
/admin/college/members	Admin
/admin/college/reports	Admin
/bulk-upload	Admin

17. Learning Hub



Learning Hub

Topic mastery heatmaps and personalized learning request management

35% Complete

Admin → `/admin/topic-mastery` or Student: `/topic-hub`

- ✓ Topic mastery heatmap
- ✓ Learning request tracking
- ⚠ Personalized learning paths (not implemented)
- ✗ Adaptive learning engine
- ✗ Topic recommendations

Path	Role
<code>/admin/topic-mastery</code>	Admin
<code>/topic-hub</code>	Student
<code>/admin/learning-requests</code>	Admin

18. Content Management



Content Management

Manage announcements, notes, cheatsheets, snippets & assignments with real-time sync

80% Complete

Admin Sidebar → Content → /admin/content → Select Type → Create → Publish

Content Types

- ✓ Announcements (with expiry, priority)
- ✓ Notes (educational references)
- ✓ Assignments (due-date linked)
- ✓ Cheatsheets (quick references)
- ✓ Snippets (code examples)

Platform Features

- ✓ Rich text editing
- ✓ File attachments (5 files, 50MB each)
- ✓ Real-time sync (WebSocket)
- ✓ Publish / draft status
- ✓ Tag-based organization
- ✓ View count tracking
- ⚠ Collaborative editing (not documented)

Path	Role
/admin/content	Admin/Instructor

Depends On

Courses

WebSocket

File Upload

19. Notifications



Notification Center

In-app notification bell and notification management center

50% Complete

Header Bell Icon

 →

Notification Dropdown

 →

/notifications (Full Center)

✓ Notification bell component

✓ Notification center page

⚠ Real-time notifications (WS partial)

⚠ Email notifications (infra ready)

✗ Notification scheduling

✗ Notification preferences

Path	Role
/notifications	All

20. Certificates



Certificates

Token-based certificate generation and verification

40% Complete

Course Completion → Generate Token → /certificate/:type/:token

- ✓ Certificate generation
- ✓ Token-based access & verification
- ⚠ Multiple certificate types (partial)
- ✗ Blockchain certificate integration
- ✗ Custom certificate templates

Path	Role
/certificate/:type/:token	Public

Depends On

Courses Users

21. Dashboard & Analytics



Dashboard & Analytics

Role-based dashboards with system-wide analytics and quick stats

65% Complete

Login → /dashboard (Role-adaptive)

- ✓ Role-based dashboard content
- ✓ Quick statistics
- ✓ Enrolled courses display
- ✓ Recent activity
- ⚠ Advanced analytics (per module)
- ✗ Custom dashboard widgets

Path	Role
/dashboard	All
/student-reports	Admin
/admin/college/reports	Admin

22. System Administration



System Administration

Feature flags, bulk operations, audit logging and system health

70% Complete

Admin Sidebar → Settings → /student-features or /bulk-upload

- ✓ Feature flag management per tenant
- ✓ Bulk user upload (CSV)
- ✓ Bulk data operations
- ⚠ System health monitoring (not documented)
- ⚠ Audit logging (partial)

Path	Role
/student-features	Admin
/bulk-upload	Admin



API Overview

Base URL: `https://<your-domain>/api/v1` · Auth: Bearer <JWT>

Module	Endpoint Group	Auth Required
Auth	<code>/auth/*</code>	Public (some)
Users	<code>/users/*</code>	JWT
Roles	<code>/roles/*</code>	JWT + manage_roles
Tenants	<code>/tenants/*</code>	JWT + manage_tenants
Courses	<code>/courses/*</code>	JWT
Batches	<code>/batches/*</code>	JWT
Attendance	<code>/attendance/*</code>	JWT + mark_attendance
Quizzes	<code>/quizzes/*</code>	JWT
Assignments	<code>/assignments/*</code>	JWT
Leads	<code>/leads/*</code>	JWT + create_leads
Public Lead Form	<code>/public/form/:tenantSlug</code>	Public (rate-limited)
Meta Webhook	<code>/meta-leads/webhook</code>	Public (verify token)
Interviews	<code>/mock-interviews/*</code> · <code>/interview-templates/*</code>	JWT
Content	<code>/content/*</code>	JWT
Notifications	<code>/notifications/*</code>	JWT
Dashboard	<code>/dashboard/*</code>	JWT



Frontend

- ✓ React 18 (TypeScript)
- ✓ React Router v6
- ✓ Context API (auth, tenant)
- ✓ Axios (API client)
- ✓ Tailwind CSS

Backend

- ✓ Node.js + Express (TypeScript)
- ✓ MongoDB + Mongoose
- ✓ JWT Authentication
- ✓ Socket.io (WebSocket)
- ✓ Multer (file upload)
- ✓ Nodemailer (email)

Infrastructure & Integrations

- ✓ Docker + Docker Compose
- ✓ Nginx reverse proxy
- ✓ GitHub Actions CI/CD
- ✓ Meta/Facebook Webhook
- ✓ Google Sheets API
- ⚠ OpenAI GPT-4 (env-gated)

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